

Denotational Semantics

An **expr ctx** $C^e \triangleq \cdot | E +_s C^e | C^e +_s E | \dots$ where $\cdot$ is the **ctx hole** (e.g. $C^e[\cdot] = \cdot +_s 2$). **Ctx application** is done by filling the hole $C^e[E]$, defined recursively as:

$$\begin{aligned} (\cdot)[E] &\triangleq E \\ (C^e +_s E')[E] &\triangleq C^e[E] +_s E' \\ (E' +_s C^e)[E] &\triangleq E' +_s C^e[E] \end{aligned}$$

This allows to combine EXP.L and EXP.R:

$$\text{EXP.E} \frac{\langle E, s \rangle \rightarrow_e \langle E', s' \rangle}{\langle C^e[E], s \rangle \rightarrow_e \langle C^e[E'], s' \rangle}$$

Weak contextual equivalence means $E_1 \sim E_2 \triangleq \forall n \in \mathbb{N}. [E_1 \rightarrow^* n \Leftrightarrow E_2 \rightarrow^* n]$, **contextual equivalence** means $E_1 \equiv E_2 \triangleq \forall C^e[\cdot]. [C^e[E_1] \sim C^e[E_2]]$.

Denotational semantics describe the *meaning* of a prog. **Interpretation** $\llbracket _ \rrbracket \subseteq \text{SimpExp} \times \mathbb{N}$ over a **semantic domain** $\mathbb{N}$:

$$\begin{aligned} \llbracket n \rrbracket &\triangleq n \\ \llbracket E_1 +_s E_2 \rrbracket &\triangleq \llbracket E_1 \rrbracket + \llbracket E_2 \rrbracket \end{aligned}$$

We can now prove properties on $+_s$. Also, we can define a *func between meanings*: $\forall C^e. \exists f \subseteq \mathbb{N}^2. [\llbracket C^e[E] \rrbracket = f(\llbracket E \rrbracket)]$. Consequently, $\forall C^e. \forall E_1, E_2. [\llbracket E_1 \rrbracket = \llbracket E_2 \rrbracket \implies \llbracket C^e[E_1] \rrbracket = \llbracket C^e[E_2] \rrbracket]$, and hence $\llbracket E \rrbracket = n \Leftrightarrow E \Downarrow n$. Also, $E_1 \equiv E_2 \Leftrightarrow \llbracket E_1 \rrbracket = \llbracket E_2 \rrbracket$.

Let Σ be the set of all states, and $\Sigma_\perp = \Sigma \cup \{\perp\}$. The **semantic domain of commands** is given by the set of **state transformers** $\mathcal{S}_c \triangleq [\Sigma \rightarrow \Sigma_\perp]$, a set of total functions that take an initial state, and return either a final state of $\perp$. Similarly, $\mathcal{S}_e \triangleq [\Sigma \rightarrow \mathbb{N}_\perp]$ and $\mathcal{S}_b \triangleq [\Sigma \rightarrow \mathbb{B}_\perp]$.

$\llbracket _ \rrbracket^e : \text{Exp} \rightarrow \mathcal{S}_e$ is defined as:

$$\begin{aligned} \llbracket n \rrbracket^e(s) &\triangleq n \\ \llbracket x \rrbracket^e(s) &\triangleq se_{lookup}(x)(s) \\ \llbracket E_1 +_s E_2 \rrbracket^e(s) &\triangleq se_{plus}(\llbracket E_1 \rrbracket^e, \llbracket E_2 \rrbracket^e)(s) \end{aligned}$$

$$\text{Lookup} \quad se_{lookup}(x)(s) = \begin{cases} s(x) & s(x) \downarrow \\ \perp & \text{o.w.} \end{cases}$$

$$\text{Addition} \quad se_{plus}(e_1, e_2)(s) = \begin{cases} e_1(s) + e_2(s) & e_1(s) \downarrow \wedge e_2(s) \downarrow \\ \perp & \text{o.w.} \end{cases}$$